SYSTEMS ENGINEERING / DECISION REFERENCE

ARCTIC UAV DELIVERY SYSTEM

Technical decision tree • aircraft, charging, landing and support equipment

PURPOSE Select a consistent technical configuration for emergency medicine delivery to ships on the Northern Sea Route. This is a decision aid, not an approved procurement or flight configuration.

Fixed design basis

Mission / constraint	Working requirement
Delivery concept	Vertical landing on ship deck; crew unloads medicine. No winch.
Energy architecture	Recharge or battery replacement on the ship and intermediate stations.
Payload and airframe	Up to 3 kg per delivery; MTOW at most 30 kg.
Route and response	Ships up to 300 km from base; request-to-handover target: 4 h.
Arctic operations	Ambient −40 to +20 °C; target winds 15 m/s steady, 20 m/s gust.
Medicine protection	Target +2 to +8 °C for 6 h; no freezing of temperature-sensitive products.

Fast choice: four architecture branches

IF the mission looks like…	THEN start from…	WHY
S1 Short-range / no station	Wingcopter 198; short direct legs	Lower route infrastructure; requires ship-landing and cold-weather validation.
S2 Long individual leg	FIXAR 025	Longer advertised range and payload headroom; speed and ambient-temperature gaps.
S3 2+ intermediate stations	Short-leg network: Wingcopter / FIXAR	Segmented range can reduce aircraft range demand; adds charging and site cost.
S4 Strict −40 °C / strong-wind ship landing	Arctic-qualified custom/OEM program	No reviewed off-the-shelf platform proven to meet the full combination.

IMPORTANT No candidate is approved for the full 300 km / 4 h / −40 °C / moving-deck envelope. Distances below are screening bands, not guaranteed payload ranges.

01 Aircraft selection tree

Decision rule: validate each flight leg with payload, wind, energy reserve, and deck-landing envelope before selecting an aircraft.

IF…	THEN…	SHORT REASON
A leg is very short (roughly ≤20–30 km) and cold/wind targets are relaxed	DJI Matrice 400 (conditional)	Simple multirotor option; the manufacturer rates only −20 °C and 12 m/s for takeoff/landing [4].
A leg is roughly ≤70 km with compact payload	Wingcopter 198 Cargo [2]	Up to 92 km advertised, 4.7 kg payload, swappable batteries; mission reserve unverified.
A compact landing solution matters more than payload headroom	RigiTech Eiger [3]	Vision-based precision landing; up to 80 km one-way at 3 kg in 2023 sheet [3b]; box size may not fit.
A leg is roughly 70–200 km	FIXAR 025 [1]	Up to 300 km claimed, 10 kg payload, 25 kg MTOW; confirm leg-specific actual range.
A leg is 200–300 km or more	FIXAR + extra station OR faster/special platform	Published 300 km is not a cold-weather, loaded, wind-reserve guarantee.
Strict −40 °C + moving ship + 20 m/s gust all mandatory	No commercial winner yet	Request OEM qualification or change mission constraints; heating alone is not proof.

Candidate comparison

Platform	MTOW / payload	Published capability	Screening concern
FIXAR 025 [1]	25 / 10 kg	Up to 300 km; 57.6 km/h cruise	−30 °C minimum; ship touchdown unproven.
Wingcopter 198 Cargo [2]	25 / 4.7 kg	Up to 92 km; hot-swap	Operating range for −40 °C unverified.
RigiTech Eiger [3]	21 / 3 kg (2023)	Up to 80 km at 3 kg; 100 km max listed	Cargo bay 45×16×21 cm; not ship-qualified.
DJI Matrice 400 [4]	15.8 / 6 kg	Up to 49 km no-wind test	−20 °C and 12 m/s takeoff/landing.

DECISION PRIORITY Always reject a platform failing a hard limit. Treat product maximum range as a preliminary screening figure, not a committed Arctic route length.

02 Charging-station count → viable leg length

Illustrative straight-line routing only. For 300 km, each station creates one additional flight leg.

Stations before ship	Ideal km per leg	Initial platform branch	Why / caution
0	300	FIXAR only as research candidate	No reserve shown at advertised maximum.
1	150	FIXAR 025	One-stop configuration; test schedule and recharge turnaround.
2	100	FIXAR 025	Wingcopter 92 km maximum is insufficient.
3	75	Wingcopter or FIXAR	Wingcopter plausible only after payload/wind/reserve check.
4	60	Wingcopter / FIXAR	More sites and stops; range demand decreases.

For a 200 km route

Stations	Ideal leg	Starting point
0	200 km	FIXAR (range analysis required)
1	100 km	FIXAR; shorter-range UAVs still lack margin
2	66.7 km	Wingcopter becomes a candidate
3	50 km	Both Wingcopter and FIXAR are candidates

Time gate: does the mission fit 4 hours?

Allowed flight + en-route charging time = 240 − 45 min dispatch − 5 min handover = 190 min.

Example for 300 km	Simple estimate	A4 result
FIXAR at 57.6 km/h, one 15-min stop	312.5 + 15 + 45 + 5 = 377.5 min ≈ 6 h 18 min	FAIL
Wingcopter at 90 km/h, three 15-min stops	200 + 45 + 45 + 5 = 295 min = 4 h 55 min	FAIL

RULE More charging stations do not automatically improve delivery time. If 4 h is fixed, shorten the route from a forward base, reduce ground time, or validate a faster platform.

These are illustrative calculations with assumed speed and stop duration. Published performance metrics may correspond to different test conditions and must not be combined as a guaranteed mission profile.

03 Ship-deck landing tree

The aircraft must land on a moving, potentially icy surface. Standard autonomous landing is not equivalent to validated maritime landing.

IF…	THEN choose…	WHY / verification
Vessel is stationary, deck motion small	GNSS/RTK + down-facing camera + illuminated fiducial + range sensor	Simplest concept; still needs field testing.
Vessel is underway / has roll, pitch or heave	Everything above + ship GNSS/INS + time sync + deck-motion prediction	Relative navigation; precise timing required.
Landing is planned in darkness	Illuminated AprilTag / ArUco plus protected camera	Visible target despite polar night, subject to snow/fog.
Visual target is obscured	Radar/range-based fallback, or abort landing	Do not assume camera-only approach works in whiteout.
Deck motion exceeds certified landing envelope	Abort / divert / wait	Safety gate overrides delivery schedule.
Touchdown completed on moving deck	Mechanical capture / tie-down + protected charging bay	Avoid slide/overturn; keep personnel clear.

Equipment alternatives

Purpose	Option A	Option B / design concern
Ship motion	SBG Ellipse-D RTK INS [5]	Existing vessel GNSS/INS if precision, interface and sync are adequate.
Landing reference	Illuminated AprilTag + global-shutter camera	UWB beacon / other relative navigation as R&D alternative.
Height	Downward LiDAR	Radar altimeter when optical performance is degraded.
Deck capture	Guided mechanical clamp / tie-down	Motion-compensating landing pad; OEM geometry compatibility required.

GATE The preliminary 5 × 5 m clear deck, ±5° roll, ±0.5 m heave and ≤1 m landing error require a ship-specific hazard and tolerance analysis. The old 5 × 5 m figure was originally discussed before switching from winch transfer to landing.

04 Energy and charging tree

Charging happens on ships and intermediate stations; battery thermal limits and OEM approval are mandatory.

IF…	THEN…	WHY
Staff present and turnaround ≤30 min	OEM battery swap + precharged packs	Faster than a full recharge; pack inventory required.
No strict turnaround	OEM charger on site	Mechanically simpler, but mission time grows.
No staff at station	Automated dock + contact/connector charging	Additional development: alignment, ice, heating and fail-safe.
Ship has stable power	Ship supply → conditioned AC/DC → OEM charger	Use onboard electrical infrastructure with protection and isolation.
Remote station has no reliable mains	Generator + buffer battery + inverter + OEM charger	Autonomous operation; more maintenance and logistics.
Ambient air is −40 °C	Heated battery store + OEM-approved warm charging	Do not charge lithium packs outside approved limits.
Operator must reuse aircraft immediately	Swap on station; charge removed pack off-aircraft	Separates aircraft service time from charging time.

What belongs to the charging station?

Core equipment	Conditional additions
Landing pad; aircraft securement; OEM charge/swap equipment; protected power distribution; fire response; communications	Heated storage cabinet; UPS/buffer battery; diesel generator; automated docking mechanism; local weather station

COMPATIBILITY FIXAR 025 battery is listed as 44.4 V, 54 Ah, but its mission-specific recharge time, swappability and approved charger must be confirmed with FIXAR [1]. Do not substitute an inverter/UPS for the OEM aviation charger.

05 Medical payload tree

All variants must preserve the medicine, fit the aircraft and keep traceable handling records.

IF…	THEN…	WHY
Passive insulation survives 6 h at −40 °C	Insulated hard case + phase-change material (PCM)	Lowest aircraft electrical demand; confirm by cold-chamber testing.
Passive insulation fails thermal tests	Heated insulated case + closed-loop control	Adds heat but consumes energy and increases mass.
Cargo contains freeze-sensitive products	PCM conditioning + no direct contact with frozen packs	Prevents local freezing despite an insulated shell.
Temperature must be documented	Independent logger (e.g., LogTag TRIX-8) + live sensor	Records chain of custody and excursions.
Fragile ampoules or vials	Shock-isolating insert + impact recorder	Needed to demonstrate target impact ≤10 g.
Total packed mass exceeds 3 kg	Rework packaging OR change payload requirement	Aircraft payload includes box, PCM, sensors and attachment.

Cargo module minimum BOM

Required by design	Optional / depends on tests
Rigid insulated box, PCM/thermal insert, temperature sensor, independent logger, sealed label, shock-protected holder, ship handover identifier	Powered heater + controller, duplicate logger, active tracker, shock logger, lockable lid

MASS DEFINITION Confirm whether “3 kg payload” is the total packaged shipment or just the medicine. RigiTech Eiger has no headroom if 3 kg is only the medicine [3].

06 Communication and navigation tree

Select links for route planning, flight control, tracking and final approach separately.

IF…	THEN…	WHY / LIMITATION
Reliable terrestrial network along route	Cellular/LTE telemetry + OEM radio	Frequent status updates; coverage must be surveyed.
Direct radio line-of-sight to station	OEM long-range RF telemetry	Independent from cellular network within validated coverage.
No terrestrial coverage	Satellite backup / satellite IP	Needed for remote Arctic corridors; antenna, latency and cost matter.
Only sparse alerts are required	Iridium 9603 short-message modem [6]	Lightweight −40 °C device; not proof of guaranteed 10 s updates.
Continuous broadband data required	Approved airborne satellite IP terminal	Higher bandwidth at greater weight, power and integration burden.
Need future ship position	Ship GNSS / AIS + motion prediction	Target is moving; AIS alone is not a precision landing signal.
Need final deck-relative position	RTK + INS + camera + marker (+ ranging)	Fuses global position and local deck motion.

Two navigation modes

Cruise to vessel	Final approach / touchdown
GNSS/INS, flight controller, forecast winds, waypoint mission, satellite/ground updates	Vessel position, accurate clock sync, relative camera/marker tracking, deck motion estimation, altimeter, abort criterion

G1 / G2 The design targets are ≥95% route communication coverage and ≥1 telemetry update every 10 seconds. No single link above is currently verified to deliver both across the whole Northern Sea Route.

07 Arctic weather and protection tree

Distinguish ambient conditions from the rated operating temperature of each device. A sensor rated −40 °C does not make an aircraft rated −40 °C.

IF we need…	CHOOSE / compare	JUSTIFICATION
Wind, pressure, temperature, humidity	Vaisala WXT536 [7] vs existing ship weather system	WXT536 operates down to −52 °C; heated variant and exposed-site verification needed.
Visibility in fog / snowfall	Vaisala PWD22 [8]	Low-visibility detection; test icing, mounting, maintenance.
Cloud base / vertical visibility	Vaisala CL31 [9]	Laser ceilometer for flight planning; optional if external station provides equivalent data.
Wind conditions from surface to 500 m	Forecast vertical profile + supplementary observations	Ground sensors cannot measure every altitude by themselves.
Polar-night landing	Illuminated landing target + protected optics	Camera-only target tracking needs visibility and contrast.
External −40 °C	Heated enclosures + winterized mechanisms + cold qualification	OEM airframe, motors, wiring, batteries and landing gear still need approval.
Icing / freezing spray	De-icing design OR weather no-go policy	No quoted off-the-shelf candidate is proven for continuous icing.

CRITICAL DISTINCTION D1 −40 °C is the EXTERNAL environment. Internal compartments may be warm, but exposed rotor/motor mechanisms, seals, charging connectors and deck equipment must still operate after a cold soak.

08 Equipment allocated by location

This is the architecture-level BOM, not a final purchase list. Conditional branches are selected only after mission trade-off.

LOCATION	BASELINE COMPONENTS	ONLY IF REQUIRED
UAV	Airframe + OEM motors/ESC; flight controller; battery/BMS; GNSS/IMU; shipment mount; aircraft health monitoring	RTK link; global-shutter landing camera; range sensor; protected heated optics; satellite radio; extra thermal controls
SHIP	Clear landing area; landing marker; protected power feed; OEM charging equipment; cargo handover process; crew communications	SBG Ellipse-D; deck capture; enclosed battery storage; autonomous charging dock; illuminated marker
INTERMEDIATE STATION	Landing pad; charger or exchange inventory; communications; power protection; safe aircraft handling	Generator; UPS; warm shelter; RTK reference; automatic docking; local weather station
MEDICAL SHORE BASE	Medicine stock and preparation; thermal packing; spare batteries; hangar; maintenance; dispatch interface	Additional aircraft, cold-storage logging, repair spares, battery conditioning room
MISSION CONTROL	Dispatch server; route and station graph; marine weather feeds; telemetry; ship tracking; audit history	Automated scheduling, SATCOM gateway, probabilistic ETA, risk scoring
MAINTENANCE & SAFETY	Inspection/checklists; flight & battery logs; spare parts, fire protection; no-go decision procedures	Cold chamber qualification, corrosion testing, ship landing trials, independent QA

Fleet sizing rule

A2 allows three concurrent missions → at least three mission-ready aircraft when flights cannot share vehicles. A spare aircraft and sufficient batteries may be necessary for availability, but both must fit budget J2 (≤15 million RUB), which has not been validated by vendor quotes.

09 Four ready-to-compare architecture packages

Each package includes all subsystems; do not mix a short-range aircraft with a long-range station assumption without rerunning the route model.

PACKAGE / IF	CONFIGURATION	ADVANTAGE → MAIN GAP
S1 SHORT-RANGE Near-base ship; few/no intermediate stops	Wingcopter 198; OEM batteries; illuminated deck marker; GNSS/INS; ship charger; passive/active thermal box	Fewer stations → battery, −40 °C, deck landing, wind still unverified.
S2 LONG-LEG One or zero intermediate stop	FIXAR 025; OEM charging; ship motion sensor; vision+range landing; satellite backup; thermal package	Long advertised range → cannot meet 300 km/4 h at 57.6 km/h cruise.
S3 CHARGING NETWORK Multiple pads; legs ~50–75 km	Wingcopter 198 or FIXAR; manual swap if staffed; weather-protected stations; distributed operations	Shorter legs → infrastructure capital, stop time and staffing grow.
S4 FULL ARCTIC −40 °C and moving ship / strong wind are hard constraints	Specialist qualified VTOL or OEM custom airframe; cold-rated exposed systems; dynamic-deck landing/capture; heated facilities	Potential full requirement coverage → development cost, tests, schedule and approvals unknown.

Selection logic in one line

PRIORITY	PASS CONDITION
1. Mission route	Every leg fits payload + headwind + reserve, and stations exist.
2. Response time	Dispatch + flights + charging/stops + ship handover ≤ A4.
3. Ship landing	Deck size, motion, wind and safety envelope are demonstrated.
4. Environment	External cold, icing and heating are validated for equipment.
5. Whole-system cost	Fleet + all stations + marine equipment + maintenance fit J-series budget.

10 Decisions and evidence required before final BOM

Current status: candidate architecture only. No product has passed all mission gates.

BLOCKING QUESTION	ENGINEERING NEXT STEP
Is 300 km the usual one-way ship distance?	Map base–station–ship graph; measure realistic segment lengths.
Is A4 = 4 h mandatory for every ship?	Resolve conflicting 5 h remark in original requirement notes; simulate route timing.
Does 3 kg include the thermal container?	Build mass budget with PCM, sensors, heating, packaging and landing hardware.
What are the deck speed, roll, pitch and heave limits?	Set dynamic landing envelope and approach/abort criteria.
Can UAV operate outside at −40 °C?	Request manufacturer cold-soak and icing evidence / OEM customization quote.
Are 15 m/s / 20 m/s valid specifically at touchdown?	Compare cruise, VTOL transition, landing and gust envelopes separately.
Is autonomous unattended recharging really required?	Choose staff-assisted swap or robotic dock; test turnaround.
Can the budget fund ≥3 aircraft plus stations?	Obtain airframe and infrastructure RFQs; estimate total lifecycle cost.

Evidence / manufacturer sources

[1] FIXAR 025 official technical sheet (07 Jul 2025)
[2] Wingcopter W198 Cargo specifications
[3] RigiTech Eiger product information
[3b] RigiTech Eiger 2023 payload-range sheet
[4] DJI Matrice 400 enterprise specifications
[5] SBG Ellipse-D GNSS/INS data
[6] Iridium 9603 module specifications
[7] Vaisala WXT536 specifications
[8] Vaisala PWD22 visibility sensor
[9] Vaisala CL31 cloud-base measurement

Source hierarchy: manufacturer specifications are used only for advertised product data; route thresholds and configurations are engineering screening assumptions; cold-weather ship operations require further validation.